Proposed AST-Route Four-Layer Multi-Objective Query Routing Framework

LAYER 1A: Input SQL & Static AST Parsing (SQLGlot)

- Workloads: 75 Factorial Queries (L1-L5) + 22 Official TPC-H (Q1-Q22)
- Zero-Execution Static AST Parsing via SQLGlot (3.476 ms mean parse time)
- Extracts Operator Topology: Depth D(q), Nodes N(q), Joins J(q), Subqueries/CTEs S(q), Windows W(q), UNNEST U(q), CASE C(q)

LAYER 1B: 9-Dimensional Feature Vector & Validated CSCI

- Composite Index: CSCI(q) = 1.5D + 2.5J + 3.0S + 3.5W + 4.0U + 1.5C
- Construct Validity: Spearman rho = 0.4935 | 10k-MC Stability rho = 0.9880
- Zone-Map Predicate Selectivity Factor: sigma(q) in (0, 1] across V1-V3
- 9D Routing Vector: x_q = [D, N, J, S, W, U, C, CSCI, sigma(q)]^T in R^9

LAYERS 2 & 3: Cross-Paradigm Analytical Cost Models & Multi-Output Cost-Regression Engine (R_GBR-9D)

- Analytical Models: PCIe Fraction Phi_PCIe(q), Effective Throughput Theta_eff (GB/s), Slot-to-Wall Proxy M_slot, FinOps Savings Delta C_1M ($)
- Multi-Output GBR Predictor f_Theta: R^9 -> R^M predicts Log-Latency ln(1+T_e), Energy E_e (Joules), and Working-Set Memory (RAM/VRAM GB)
- Out-of-Fold Predictive Fidelity (25-Seed LOSO-CV): GPU R^2 = 0.875 (Q-Error P50 = 1.053) | Polars R^2 = 0.822 (P50 = 1.095) | Energy R^2 = 0.815 (497.2 us)

Candidate 1: DuckDB (C++)

- Paradigm: In-Process CPU DBMS
- Morsel-Driven + Out-of-Core Spill
- Segment-Tree Window Framing
- SF1 Mean: 183.74 ms (6.40 J)
- TPC-H Q1-Q22 Mean: 22.82 ms
- Routed: 80% L1, Q16-Q22, SF30/100 L5

Candidate 2: Polars / DataFusion

- Paradigm: Rust Apache Arrow
- Zero-Copy SIMD & Streaming
- Polars SF1: 82.47 ms (3.17 J)
- DataFusion SF1: 96.30 ms (3.39 J)
- 102.1x Faster than Row-Store
- Routed: 80% Narrow/Med L2 Joins

Candidate 3: CUDA GPU (SIMT)

- Paradigm: RTX 2050 (4GB VRAM)
- SF1 Cold: 56.96 ms | Warm: 22.26 ms
- Multi-Scale: 1x, 4x, 12x, 40x PCIe
- Energy: 0.65 J Cold / 0.25 J Warm
- Green Perf: 1942.8 MB/J Warm
- Routed: 60% L3, 100% L4, 80% L5

Candidate 4: Google BigQuery

- Paradigm: Serverless Cloud OLAP
- Capacitor Storage + Slot Elasticity
- Clustered SF1 Mean: 745.39 ms
- Slot-to-Wall Proxy: M_slot = 6.37x
- V1 Pruning: 78.4% (1.47 – 147/1M)
- Wins SF100 L1/L4 vs. Local CPUs

LAYER 4: Pareto-Configurable Engine Dispatcher & Generalization Across Unseen Seeds, TPC-H Q1-Q22, and SF=1-100 GB

- Pareto Dispatch Rule: e_hat_alpha(q) = argmin_e [alpha*(T_hat/T_med) + (1-alpha)*(E_hat/E_med)] -> 54.92 ms (0.00% Median Regret, 34.98% Mean Regret)
- Generalization: 73.33% Top-1 Acc (47.01 ms) on Unseen Seeds | 72.73% Top-1 Acc (15.84 ms) on Official TPC-H Q1-Q22 | 4.63x-7.63x Speedup Across SF=1-100 GB